From Reactive to Predictive: A Federated Graph-Attention SAC Framework for Schedule-Driven WiFi Optimization in High-Density Transit Venues

Quang-Vinh Dang[a,1,*]

[a]*British University Vietnam, Hung Yen, Vietnam*

Abstract

High-density transit venues—such as airport terminals, railway stations, and bus interchanges—experience severe, schedule-driven congestion where passenger crowds gather predictably around arrival and departure events. Commercial WLAN controllers are largely reactive: they rebalance radio resources after congestion has already degraded service. In this work, we propose **STAG-FedSAC**, an end-to-end framework for proactive, schedule-driven WiFi radio resource management. STAG-FedSAC integrates three coordinated modules: (i) **ST-GCAT**, a schedule-conditioned spatio-temporal graph attention Transformer that predicts AP-level load up to 30 minutes ahead by cross-attending multi-step transit timetables with spatial AP interference topologies; (ii) multi-agent **Soft Actor-Critic (SAC)** agents per AP that perform proactive, multi-objective control over transmit power, channel selection, and quantized 802.11e EDCA airtime quotas under a Lagrangian airtime constraint; and (iii) **HierFed-KD**, a hierarchical federated distilla-

*Corresponding author

Email address: vinh.dq4@buv.edu.vn (Quang-Vinh Dang)

[1]ORCID: 0000-0002-3877-8024

tion protocol that combines intra-zone parameter averaging with inter-zone proximal knowledge distillation over a public reference set, which keeps per-user data and replay buffers at the AP (data minimisation, not a formal privacy guarantee) and reduces inter-zone communication.

To increase physical and protocol realism, our evaluation incorporates IEEE 802.11 MAC contention based on Bianchi's saturation model, channel-switching outage penalties, and EDCA-based airtime budgeting. Over three training seeds, the proposed configuration of STAG-FedSAC (which feeds compact schedule features, not the latent forecaster embedding, to the policy) matches the static incumbent plan SSF in reward (0.182 against 0.181, a difference within seed variation) and reduces the average user latency to 109.9 ms (11.8% below SSF at 124.6 ms); its throughput (1.32 Mbps/user), QoS satisfaction (0.689) and energy efficiency (7.47 Mbps/W) are comparable to those of SSF. Relative to the learned baselines it improves the reward by 56.9% (FedDDPG) and 68.5% (LSTM-DRL) and reduces latency by 40.5% and 44.7%; these baselines change the channel of 0.67–0.75 of the APs per interval, against 0.15 for STAG-FedSAC and 0.075 for SSF. ST-GCAT reduces the load-forecast RMSE relative to persistence by 35% overall and 39% in surge windows, and relative to the same predictor without the schedule by 31% and 33% (at the one-step horizon, simple forecasters are more accurate). HierFed-KD reduces inter-zone wide-area communication by 34× relative to weight-based hierarchical federation and 1711× relative to flat FedAvg. In our ablations, feeding the forecaster's latent embedding to the policy and the joint predictor–policy gradient did not improve control. All results are from simulation.

Keywords: WiFi resource management, Spatio-temporal graph attention, Soft Actor-Critic, Federated learning, Knowledge distillation, Transit venue networks, Deep reinforcement learning, Schedule-aware prediction

1. Introduction

Transit venues—airports, railway stations, and bus terminals—are among the most demanding environments for WiFi. At peak hours a single departure lounge can host hundreds of users streaming video, making calls, and uploading content over a handful of access points (APs) [1]. Unlike enterprise offices or university campuses, transit venues have a critical feature that a network controller can exploit: a substantial fraction of the congestion is *announced in advance*. Departure and arrival timetables indicate when and where passenger crowds will gather, with a temporal precision rarely available in other wireless settings. Commercial WLAN controllers, however, are largely *reactive*: they detect congestion after users experience degraded service and respond with local load-balancing heuristics that ignore future schedule information [2].

In this paper, we investigate how schedule information can be systematically integrated into proactive radio resource optimization. We organize the investigation around four fundamental research challenges:

1. **Schedule unawareness**: conventional load predictors treat AP load as a purely historical time series and discard timetable metadata that provides crucial lead time before an impending surge [3].
2. **Reactive radio resource control**: heuristic controllers (e.g., strongest signal first or least loaded first) and conventional deep reinforcement

learning (DRL) agents trained without future look-ahead cannot pre-configure channel assignments, transmit powers, and airtime allocations before crowds physically arrive [4].

3. **Multi-class QoS alignment**: traffic in transit hubs is heterogeneous: VoIP requires ultra-low latency, video demands high throughput, and best-effort traffic tolerates delay. This necessitates a hybrid, multi-objective action space (continuous transmit power, discrete channel choice, and per-class airtime budgeting) that standard continuous DRL architectures struggle to coordinate stably [5].
4. **Decentralized learning across multi-operator administrative domains**: in large transit terminals, distinct zones often belong to different administrative operators (e.g., airport authority, airline lounges, retail tenants) who cannot pool sensitive per-user association and traffic data. Isolated per-AP learners fail to exploit broader spatial correlations, while monolithic centralized learning violates data privacy boundaries [2].

To address these challenges, we propose **STAG-FedSAC**, an integrated framework that synergistically combines: (i) a schedule-conditioned spatio-temporal graph attention Transformer (**ST-GCAT**) that forecasts multi-horizon AP load up to 30 minutes ahead; (ii) per-AP **Soft Actor-Critic (SAC)** agents that leverage schedule context to optimize transmit power, channel selection, and quantized 802.11e EDCA airtime quotas under a Lagrangian airtime utilization constraint; and (iii) a two-level hierarchical federated learning protocol (**HierFed-KD**) that combines intra-zone quality-weighted model averaging with inter-zone proximal knowledge distillation

over a public reference dataset, eliminating raw weight exchanges across administrative boundaries. Figure 1 presents an architectural overview, and Section 3.1 details the operational roles of all entities.

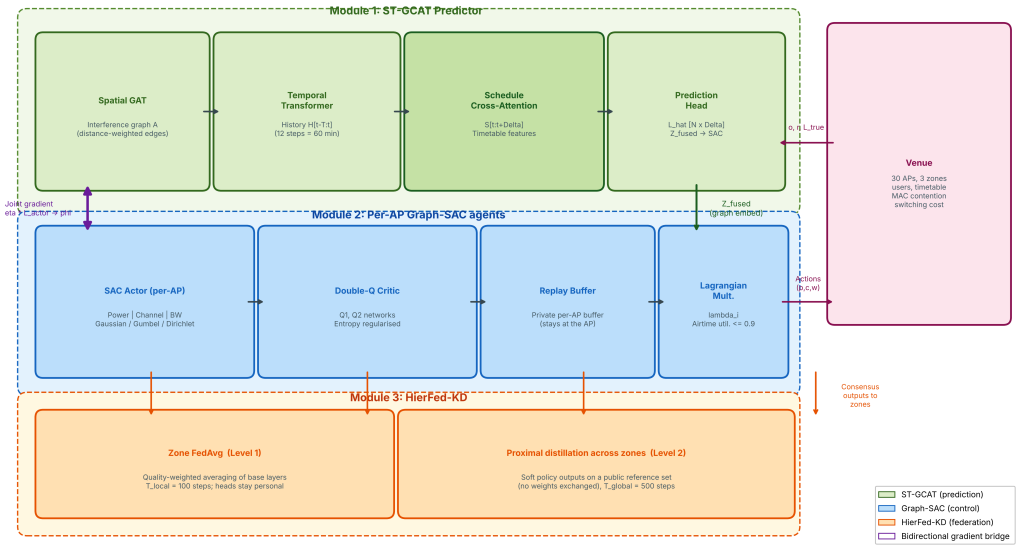


Figure 1: STAG-FedSAC architecture overview. Module 1 (ST-GCAT) forecasts AP load up to 30 minutes ahead from historical counters, the inter-AP interference graph, and scheduled transit timetable events, producing spatio-temporal representations. Module 2 (per-AP SAC agents) conditions on schedule context and local radio states to proactively select transmit power, channel assignment, and 802.11e EDCA airtime quotas under a Lagrangian airtime constraint. Module 3 (HierFed-KD) orchestrates two-level federation: quality-weighted parameter averaging within local zones and communication-efficient proximal distillation over a public reference set across administrative zones. Sensitive per-user association data and replay buffers remain strictly local to each AP.

Contributions. Our principal technical contributions are summarized as follows:

- **Schedule-Conditioned Spatio-Temporal Forecasting (ST-GCAT)**:

We design a cross-attention graph Transformer architecture that fuses transit timetable features (event arrival/departure times, passenger volumes, and event proximity) with historical AP counters and interference topologies, providing multi-step forecasts up to $\Delta = 6$ intervals (30 minutes) ahead. ST-GCAT reduces the surge-window forecast RMSE by 39% relative to persistence, by 33% relative to the same predictor without the schedule and by 24% relative to a temporal-only LSTM.

- **Proactive Multi-Objective Radio Resource Allocation**: We formulate the joint optimization of transmit power, channel selection, and multi-class QoS airtime budgeting as a constrained Markov Decision Process (CMDP). Using entropy-regularized Soft Actor-Critic with personal heads and adaptive Lagrangian multipliers, our agents reach the reward of the static incumbent plan SSF within seed variation (0.182 against 0.181) with a lower average user latency (109.9 ms, 11.8% below SSF at 124.6 ms), and outperform the learned DRL baselines (FedDDPG and LSTM-DRL) by 40.5%–44.7% in latency and over 56% in reward, with far fewer channel switches than those baselines (0.15 against 0.67–0.75 switches/AP/step).

- **HierFed-KD Multi-Operator Federation**: We develop a two-level federated learning protocol that respects administrative domain boundaries. Intra-zone APs average their representations with reward-based weighting, while inter-zone coordination is achieved via proximal knowledge distillation over soft policy outputs on a public reference set. This design provides data minimisation (not a formal privacy guarantee) and

reduces inter-zone wide-area communication by 34× relative to weight-based hierarchical federation and 1711× relative to flat FedAvg.

- **Comprehensive Evaluation with Protocol Realism**: Departing from simplistic Shannon-capacity models, we implement a more realistic network simulator incorporating IEEE 802.11 MAC contention based on Bianchi's saturation model, channel-switching outage delays, and 802.11e EDCA-based airtime quotas. Multi-seed experiments (three seeds for the main comparison), paired bootstrap confidence intervals, and systematic ablation studies give insights, for our simulator, into representation design and gradient coupling for wireless reinforcement learning.

The remainder of this paper is organized as follows: Section 2 surveys related work. Section 3 formalizes the system model and the three modules of STAG-FedSAC. Section 4 presents experimental setup, comparative evaluations, ablation studies, and robustness analyses. Section 5 discusses deployment considerations, and Section 6 concludes the paper.

2. Related Works

2.1. Spatio-Temporal Prediction for Wireless and Wi-Fi Networks

Accurate prediction of access-point (AP) load and wireless traffic is a prerequisite for proactive resource management, and spatio-temporal deep learning has become the dominant approach. Shaabanzadeh and Sánchez-González [6] evaluate a broad family of architectures (CNN, GRU, LSTM and Transformer) for spatio-temporal Wi-Fi load prediction at the individual AP

level on real campus measurements spanning 100 APs over three months, fusing spatial neighbour information only when inter-AP correlations are statistically significant.

At cellular scale, Cai et al. [7] propose DBSTGNN-Att, a dual-branch spatio-temporal graph neural network with self-attention that fuses location and point-of-interest features through an attributed k-nearest-neighbour graph, with superior short-, mid- and long-term predictions on the Milan Telecom Italia dataset. Lin et al. [8] unify normal, target-alarm and neighbour-alarm network states by transfer learning over spatio-temporal graph neural networks for multi-scenario cellular KPI prediction. Xiao et al. [9] combine a diffusion convolutional GRU with multi-head attention to capture spatial diffusion, temporal periodicity and external influence in cellular traffic.

The work closest in spirit to ours is STECA-GCN [10], which incorporates an event dimension via cross-attention in a spatio-temporal graph model and feeds the output into a DDPG-based load-balancing agent; its event features are, however, coarse binary workday/holiday labels. Among the works we reviewed, we did not find one that uses structured transit timetable data (arrival and departure times and passenger counts) as cross-attention context for Wi-Fi AP load prediction; this is the setting of our ST-GCAT predictor.

2.2. Deep Reinforcement Learning for Wireless Resource Allocation

Deep reinforcement learning has been widely applied to continuous resource-management problems in wireless networks. Yang et al. [11] identify a discounted-reward mismatch in prior wireless RL formulations and propose ARO-SAC, an average-reward Soft Actor-Critic variant for RAN slicing, with

a 15% gain over conventional SAC on the NetworkGym simulator. For energy efficiency in RIS-assisted UAV systems, Adhikari et al. [12] jointly optimise phase shifts and 3D trajectory with twin delayed DDPG, benchmarking against SAC, DDPG and double DQN. Wang et al. [13] address 6G mmWave massive MIMO-NOMA with a hybrid DRL framework that pairs dueling DQN for discrete subchannel allocation with DDPG for continuous power allocation. Pi and Ren [14] develop HMSAC, a heterogeneous-action multi-agent SAC with an embedded attention mechanism that jointly optimises channels, power levels and bandwidths for NOMA uplink in vehicular networks. Qi et al. [15] formulate a multi-QoS DRL problem for RIS-aided IoV networks that minimises age of information under vehicle-to-vehicle reliability constraints. In the works we reviewed, the environment state is a scalar or flat vector (channel gains, per-slice metrics, positions); we did not find a work that builds the RL state from the dense, spatially structured embedding of a spatio-temporal predictor, which is what our Graph-SAC state does.

2.3. Federated Learning for Wireless Networks

Azimi-Abarghouyi and Fodor [16] introduce scalable hierarchical over-the-air federated learning with a two-level cluster/global aggregation scheme and convergence bounds for heterogeneous wireless topologies. Pervej et al. [17] address bandwidth scarcity and device heterogeneity in hierarchical FL by adaptive model pruning during inter-level aggregation. At the aggregation-mechanism level, Shao et al. [18] propose Selective-FD, which selectively shares soft-label knowledge without a well-trained teacher, and Pang et al. [3] discuss how federated learning and knowledge distillation can be combined in wireless systems. Liang et al. [19] jointly optimise communication energy and

training latency in over-the-air FL with a DRL algorithm, and Wu et al. [20] study age-based client selection and resource allocation for FL over NOMA networks. Li et al. [21] use knowledge distillation for asynchronous federated reinforcement learning in IoT. The hierarchical schemes above average or prune parameters, Selective-FD is formulated for classification, and the FL-wireless co-design works use flat FedAvg-style aggregation. We did not find a scheme that combines zone-level averaging, inter-zone distillation of *policy outputs* on a public reference set and Wi-Fi resource optimisation as the target application; we do not claim that none exists.

2.4. Graph Neural Networks in Networking

Dai et al. [22] survey graph-based resource management in wireless networks (GCN, GAT, GraphSAGE, MPNN) and identify GNN-enabled DRL as a major open direction. Liu et al. [23] integrate graph attention with multi-agent RL to address partial observability and non-stationarity. Ji et al. [24] combine GraphSAGE with DRL so that low-dimensional graph embeddings serve directly as the distributed agents' state for channel selection and power control in C-V2X. Marwani and Kaddoum [25] model multi-link interference as a graph and train a GNN to solve power control and spectrum allocation jointly, and Li et al. [26] use a heterogeneous GNN with distinct node and edge types for cell-free massive MIMO power allocation.

Ji et al. [24] is architecturally the closest to ours, since GNN embeddings are the RL state; it uses a static interference-topology convolution rather than a forecast-conditioned spatio-temporal embedding, employs DQN rather than SAC, and has no federated training. We did not find prior work that passes the output of a schedule-conditioned graph forecaster into the state of a SAC

policy.

3. Methodology

This section first fixes the system model (who observes what, who decides, and what is shared), then defines the control problem, and finally describes the three modules of STAG-FedSAC. Every symbol is defined where it is first used; Table 1 collects them.

3.1. System Model and Roles

We consider a venue with N access points (APs) grouped into Z *zones* (e.g. departure hall, arrival hall, concourse). Three kinds of entities take part (Fig. 1):

- **Per-AP agent.** Each AP i is controlled by its own Soft Actor-Critic (SAC) agent: an actor π_i, two critics Q_i^1, Q_i^2 with target copies, and a private replay buffer $\mathcal{B}_i$. Agents do not share a policy by default; π_i keeps a *personal* head for the traffic of its own area. An agent runs on the AP itself or on the controller of the zone, observes only the quantities listed in Table 2, and every control interval chooses the action of *its* AP (Section 3.2).

- **Zone controller** (one per zone). Aggregates the models of the APs of its zone (Level 1 of HierFed-KD).

- **Venue controller** (one per venue). Hosts the ST-GCAT predictor, which needs AP-level load counters of all APs and the timetable; it also coordinates the zones (Level 2) and serves the public reference set used for distillation.

Time and data flow. Time advances in control intervals of 5 minutes. At the start of interval t, every AP reports its counters (Table 2) to the venue controller; ST-GCAT turns the last $T = 12$ intervals (one hour) of counters of all APs, the interference graph and the timetable of the next $\Delta = 6$ intervals into one embedding per AP; each agent then selects its action, and the reward is computed from the counters of the interval that follows.

What is federated, and why. Per-user data (association records, device identifiers, traffic mix, mobility traces) exist only to train the AP agents, and in a large venue the zones are frequently run by different parties (terminal operator, airline lounges, retail tenants) who cannot pool such data. Federated learning is therefore used to let agents benefit from each other's experience while *transitions, per-user records and replay buffers never leave the AP*; only (parts of) model parameters or, at Level 2, soft policy outputs on a public reference set are exchanged. The predictor consumes AP-level aggregates only (counts and averages, no user identifiers), which are already collected by any WLAN controller. We want to be explicit about the scope of this protection: it is *data minimisation*, not a formal privacy guarantee. We do not add differential privacy or secure aggregation, and exchanged parameters or outputs can still leak information about local data. Section 4.7 quantifies what federation buys in learning performance and communication, by comparing it with isolated learners and with a (privacy-violating) pooled-experience learner.

Table 1: Main notation.

Symbol	Meaning
N, Z, C	number of APs, zones, channels
T, Δ	history length (12 intervals) and forecast horizon (6 intervals)
$\mathbf{H}$, $\mathbf{S}$, $\mathbf{A}$	AP history, schedule tensor, interference graph
L_i^t, $\hat{L}_i^t$	normalised load of AP i and its forecast
$\mathbf{Z}_{\text{fused}}$, d_h	ST-GCAT embedding and its width (128)
$\mathbf{s}_i$, $\mathbf{a}_i = (p_i, c_i, \mathbf{w}_i)$	state and action of AP i (power, channel, airtime budget)
u_i, λ_i	airtime utilisation and Lagrange multiplier of AP i
$\alpha, \beta, \gamma, \delta, \delta_p$	reward weights
η, κ	joint-gradient weight and embedding scale
μ, T_{local}, T_{global}	proximal weight and the two federation periods
M, $\mathcal{S}_{\text{ref}}$	size and set of public reference states
t_{sw}	channel-switch outage

Table 2: Information available to the framework at control interval t (per AP unless stated).

Input	Content
Local features $\mathbf{h}_i^t$	normalised user load L_i, throughput, mean latency, airtime utilisation u_i, numbe
History $\mathbf{H}$	$\mathbf{h}_i^{t-T+1}, \ldots, \mathbf{h}_i^t$ of all APs, $T = 12$ (oldest frame repeated at episode start)
Interference graph $\mathbf{A}$	weight $a_{ij} = 1 - d_{ij}/80\,\text{m}$ for $d_{ij} < 80$ m, else 0
QoS mix $\mathbf{q}_i^t$	fractions of VoIP / video / best-effort users
Radio context $\mathbf{r}_i^t$	own channel (one-hot), users and co-channel APs sensed in its collision domain (
Schedule $\mathbf{S}^{t:t+\Delta}$	8 features per AP and future interval derived from the timetable (Table 3)

Table 3: Schedule features $\mathbf{S}_i^\tau$ of AP i for a future interval τ (events of the zone of AP i; normalising constants in brackets). A timetable event is a departure or arrival with a time t_e and a number of passengers n_e.

#	Feature
1, 2	departure / arrival surge $\sum_e n_e \exp\big(-(t_e-\tau)^2/2\sigma^2\big)$, $\sigma = 2$ intervals [300]
3	1 if an event lies within ± 3 intervals of τ
4	intervals until the next departure (0 if none) [12]
5	intervals since the last arrival (0 if none) [12]
6	occupancy forecast $\min(1, (f_1 + f_2)/300)$
7	number of events of the zone in the next Δ intervals [3]
8	zone index / 2

3.2. Control Problem

Action. At each interval AP i chooses $\mathbf{a}_i^t = (p_i, c_i, \mathbf{w}_i)$: transmit power $p_i \in [P_{\min}, P_{\max}] = [5, 23]$ dBm, operating channel $c_i \in \{1, \ldots, C\}$ ($C = 11$ orthogonal 20 MHz channels), and an airtime budget $\mathbf{w}_i \in \Delta^3$ over the three QoS classes (VoIP, video, best-effort). How c_i and $\mathbf{w}_i$ are realised on commodity hardware, and what they cost, is described in Section 3.7.

Reward. The reward of AP i is

$$r_i^t = \alpha\, r_i^{\mathrm{tp}} + \beta\, r_i^{\mathrm{lat}} + \gamma\, r_i^{\mathrm{fair}} + \delta\, r_i^{\mathrm{qos}} + \delta_p\, b_i^{\mathrm{pred}}, \tag{1}$$

with every term on a unit scale:

$$r_i^{\mathrm{tp}} = \min\bigl(1,\, X_i / X_{\mathrm{ref}}\bigr), \qquad X_i = \text{aggregate throughput of AP } i \text{ [Mbps]},\ X_{\mathrm{r}}$$

$$r_i^{\mathrm{lat}} = -\min\bigl(1,\, \ell_i / \ell_{\mathrm{ref}}\bigr), \qquad \ell_i = \text{mean user latency [ms]},\ \ell_{\mathrm{ref}} = 100,$$

$$r_i^{\mathrm{fair}} = \frac{\bigl(\sum_u \varsigma_u\bigr)^2}{n_i \sum_u \varsigma_u^2}, \qquad \varsigma_u = \text{throughput of user } u \,/\, \text{its offered rate},$$

$$r_i^{\mathrm{qos}} = -\frac{\bigl|\{u :\ u \text{ violates its class requirement}\}\bigr|}{n_i}, \qquad \text{requirement: SINR} \geq \theta_c,\ \text{latency} \leq \ell_c^{\mathrm{SLA}},\ \varsigma_u$$

$$b_i^{\mathrm{pred}} = -\bigl|\hat{L}_i^{t+1} - L_i^{t+1}\bigr|, \qquad \text{(training signal only; not part of the evaluat}$$

The weights $\alpha = 0.40$, $\beta = 0.25$, $\gamma = 0.20$, $\delta = 0.05$, $\delta_p = 0.10$ sum to one and encode an operator priority order (throughput, then latency, then fairness; the QoS-violation term is small because violations already lower the throughput and latency terms, and the prediction term is a small auxiliary shaping signal). They are a design choice, not the output of a derivation: we do not claim they are optimal, and Section 4.9 reports how conclusions change under three other weightings. SAC is trained on the reward multiplied by a constant scale $R_{\mathrm{s}} = 10$ that matches the reward magnitude to the entropy term; *evaluation always uses the unscaled environment reward of Eq.* (1) *without* b^{pred}, so that methods without a predictor are scored identically.

Constraint. Let $\rho_i^{\mathrm{need}} = \sum_{u \in \mathcal{U}_i} d_u / R_u$ be the airtime AP i needs to carry the offered rates d_u at the PHY rates R_u of its users, and ρ_i^{avail} the airtime it actually obtains (its share of the collision domain times the MAC efficiency, Section 4.1). We require the *airtime utilisation* $u_i^t = \rho_i^{\mathrm{need}} / \rho_i^{\mathrm{avail}} \leq u_{\max} = 0.9$. In contrast to the number of associated users, u_i *can be controlled* by power, channel and airtime split, which is what makes it a meaningful constraint.

It is enforced by an adaptive multiplier,

$$\lambda_i^{t+1} = \mathrm{clip}\Big(\lambda_i^t + \rho_\lambda \, \min\big(u_i^t - u_{\max},\, 0.3\big),\, 0,\, \lambda_{\max}\Big), \qquad \rho_\lambda = 0.01,\ \lambda_{\max} = 1, \tag{2}$$

and the penalty $-\lambda_i \min\big(\max(u_i - u_{\max}, 0), 0.3\big)$ is added to the reward the agent learns from (it is not part of the evaluation reward). The bound on λ_i and the clipped violation keep the penalty on the scale of the reward when an AP is persistently overloaded and the constraint cannot be met.

3.3. Module 1: ST-GCAT Predictor

ST-GCAT forecasts the normalised load $L_i^{t+1:t+\Delta}$ of every AP ($L_i = \min(1, \mathrm{users}_i/60)$) and produces the embedding $\mathbf{Z}_{\mathrm{fused}}$ used as part of the agents' state. Table 4 gives the exact sizes of all learned maps.

Table 4: ST-GCAT layer dimensions ($d_h = 128$, $K = 4$ heads, $d_k = d_h/K = 32$, $F = 5$ input features, $D_s = 8$ schedule features, $\Delta = 6$).

Map	**Shape**	**Role**
$\mathbf{W}_{\mathrm{in}}$, $\mathbf{W}_s$	$F \times d_h$, $D_s \times d_h$	input projections of $\mathbf{H}$ and $\mathbf{S}$ (the
$\mathbf{W}^{\mathrm{sp}}$	$d_h \times d_h$	GAT linear map, output split into
$\mathbf{a}^k = [\mathbf{a}_l^k; \mathbf{a}_r^k]$, $k = 1..K$	$2d_k = 64$ per head	GAT attention vector of head k (c
Temporal Transformer	2 layers, K heads, feed-forward width $4d_h$	self-attention over the T time step
Cross-attention	K heads, $d_h = 128$	queries from the temporal summar
Prediction head	$d_h \to 64 \to 1$, ReLU, sigmoid	maps each $\mathbf{Z}_{i,k}$ to a load in $[0, 1]$

Spatial GAT. For every time step of the history, node features are mixed over the interference graph. For head k and nodes i, j (with $\mathbf{g}_i = \mathbf{W}^{\mathrm{sp}}\mathbf{x}_i$ and

$\mathbf{g}_i^k \in \mathbb{R}^{d_k}$ its k-th block),

$$e_{ij}^k = \mathrm{LeakyReLU}\big(\mathbf{a}^{k\top}[\mathbf{g}_i^k \,\|\, \mathbf{g}_j^k]\big) + 0.1\, a_{ij} = \mathrm{LeakyReLU}\big(\mathbf{a}_l^{k\top}\mathbf{g}_i^k + \mathbf{a}_r^{k\top}\mathbf{g}_j^k\big) + 0.1\, a_{ij}, \quad \alpha_{ij}^k = \frac{\exp e_{ij}^k}{\sum_{m\in\mathcal{N}_i\cup\{i\}} e} \tag{3}$$

and $\mathbf{h}_i^{\mathrm{sp}} = \mathrm{ELU}\big(\|_k \sum_j \alpha_{ij}^k \mathbf{g}_j^k\big) \in \mathbb{R}^{d_h}$, where $\mathcal{N}_i = \{j : a_{ij} > 0\}$ and $\|$ is concatenation over the heads.

Temporal Transformer and schedule cross-attention. For each AP, a Transformer encoder runs over the T spatially mixed frames and its last output $\mathbf{u}_i \in \mathbb{R}^{d_h}$ summarises the history. The summary is repeated for the Δ forecast steps and then *queries* the schedule through multi-head cross-attention,

$$\mathbf{Z}_i = \mathrm{LN}\big(\mathbf{U}_i + \mathrm{CrossAttn}(Q = \mathbf{U}_i,\, K = \mathbf{S}_i,\, V = \mathbf{S}_i)\big) \text{ followed by a residual feed-forward block,} \tag{4}$$

with $\mathbf{U}_i \in \mathbb{R}^{\Delta\times d_h}$ and $\mathbf{S}_i$ projected to $\mathbb{R}^{\Delta\times d_h}$. The embedding $\mathbf{Z}_{\mathrm{fused}} = [\mathbf{Z}_1, \ldots, \mathbf{Z}_N] \in \mathbb{R}^{N\times\Delta\times d_h}$ is passed to the agents. Because the summary $\mathbf{U}_i$ is the same for all Δ steps, the horizon-specific information in $\mathbf{Z}_i$ comes from the schedule; a schedule-free variant (ablation A2, Section 4.6) therefore has no horizon-specific signal other than the positional structure learned by the head.

Training targets. The targets of the predictor are the *true* loads L_i^{t+1+k}, $k = 0..\Delta-1$. A sample $(\mathbf{H}, \mathbf{S}, \text{targets})$ is stored when the corresponding future intervals have been observed (at the end of the episode; steps beyond the end of the episode are masked), and the predictor is trained on minibatches of 16 stored samples.

3.4. Module 2: Graph-SAC Agents

State. The state of agent i concatenates its local features, its slice of the ST-GCAT embedding, its QoS mix, its next-interval schedule features and its radio context:

$$\mathbf{s}_i^t = \left[\mathbf{h}_i^t,\ \kappa\,\mathrm{vec}(\mathbf{Z}_{\mathrm{fused},i}^t),\ \mathbf{q}_i^t,\ \mathbf{S}_i^{t+1},\ \mathbf{r}_i^t\right] \in \mathbb{R}^{5+768+3+8+13=797}. \tag{5}$$

The scalar $\kappa = 0.03$ scales the embedding block. The layer-normalised embedding has 768 entries of unit scale (norm ≈ 28), whereas the other 29 inputs have a combined norm of about 3; unscaled, the embedding dominates the first layer of the actor and critic. We chose κ among $\{1, 0.125, 0.03\}$ on validation episodes of one seed (seeds disjoint from the test seeds); larger values gave lower validation reward. This choice is part of the method and its consequences for the contribution of the embedding are discussed in Section 4.6. In the experiments, the configuration called STAG-FedSAC (Proposed) sets the embedding block to zero (the policy then uses the schedule features $\mathbf{S}_i^{t+1}$ and the other inputs) and the configurations that use it ($\kappa = 0.03$) are labelled "Latent Emb."; this choice was made after the ablation results were available.

Actor. The actor has a federated *base* (two layers of 256 units) and a personal *head* (128 units) followed by three output heads: power (Gaussian, squashed by a sigmoid to $[P_{\min}, P_{\max}]$ with the corresponding log-density correction), channel (straight-through Gumbel-softmax over C categories with temperature annealed from 2.0 to 0.5, so that the one-hot vector that is executed is the one stored and given to the critic) and airtime budget (Dirichlet with concentrations $\mathrm{softplus}(\cdot) + 1$). The critics are double-Q networks with two

hidden layers of 256 units. The updates are the standard SAC updates,

$$\mathcal{L}_Q = \mathbb{E}\Big[\big(Q(s,a) - r - \gamma(\min_j \bar{Q}_j(s',a') - \alpha_{\text{ent}} \log \pi(a'|s'))\big)^2\Big], \tag{6}$$

$$\mathcal{L}_\pi = \mathbb{E}\big[\alpha_{\text{ent}} \log \pi(a|s) - \min_j Q_j(s,a)\big] + \tfrac{\mu}{2}\|\theta_{\text{base}} - \theta_{\text{anchor}}\|^2, \tag{7}$$

where α_{ent} is the entropy temperature (initialised to 0.05, automatically tuned towards a target entropy of -3) and the last term is the FedProx proximal term towards the last received base (active only in the two hierarchical variants). One gradient update of all agents is made after every environment step (update-to-data ratio one).

Starting from the incumbent configuration. In an operating venue the network already has a configuration (channel plan, power, equal airtime). All learned agents, including the learned baselines, are initialised close to it: the channel head is biased towards the AP's incumbent channel (round-robin plan), power starts at 14 dBm, the airtime split at $(frac13, \frac{1}{3}, \frac{1}{3})$, and the output layers are scaled down so that the initial policy depends little on the state. Exploration during the warm-up episodes is the noise of this initial stochastic policy around the incumbent configuration, not uniformly random actions; learning can then only depart from the incumbent where it finds an improvement. The one exception to the parametrisation is the single shared agent of LSTM-DRL, which has no AP-specific parameters and therefore outputs its channel as an offset from the AP's incumbent channel.

3.5. Bidirectional Joint Gradient

The predictor is trained on both the forecasting error and the actors' objective,

$$\mathcal{L}_{\text{pred}} = \underbrace{\frac{1}{N\Delta}\sum_{i,k}\left(\hat{L}_{i,k} - L^{*}_{i,k}\right)^{2}}_{\text{forecast error}} + \eta\,\frac{\mathcal{L}_{\pi}(\phi)}{\overline{|Q|}+1}, \qquad \eta = 0.10, \tag{8}$$

where $\mathcal{L}_{\pi}(\phi)$ is Eq. (7) evaluated on states rebuilt from the *current* predictor embedding of stored samples (so the gradient $\partial\mathcal{L}_{\pi}/\partial\phi$ reaches ST-GCAT through $\mathbf{Z}_{\text{fused}}$), and $\overline{|Q|}$ is the mean magnitude of the critic values in the batch (treated as a constant). Normalising by $\overline{|Q|}+1$ makes η independent of the reward scale; without it, the actor term dominates the gradient of the predictor. Only ϕ is updated in this step; the actor and critic parameters are updated by Eqs. (6)–(7). With $\eta = 0$ (ablation A1) the predictor is still trained, on the forecast error alone. States stored in the replay buffers contain the embedding of the time of storage; they are not recomputed when the predictor changes.

3.6. Module 3: HierFed-KD

Only the actor base layers ($797\cdot256 + 256 + 256\cdot256 + 256 = 270{,}080$ parameters) are federated; heads and critics stay local.

Level 1: zone aggregation. Every $T_{\text{local}} = 100$ environment steps the controller of zone z replaces the base of each AP by the quality-weighted average

$$\theta_{z}^{\text{zone}} = \sum_{i\in\mathcal{Z}_z} w_i\,\theta_i, \qquad w_i = \frac{\exp(\bar{r}_i/\tau_w)}{\sum_{j\in\mathcal{Z}_z}\exp(\bar{r}_j/\tau_w)}, \quad \tau_w = 0.1, \tag{9}$$

where $\bar{r}_i$ is the exponential moving average of the (unscaled) environment reward of AP i.

Level 2: proximal federated distillation across zones. Every $T_{\text{global}} = 500$ steps the zones do *not* exchange weights. Instead, all agents evaluate their actor on a public reference set $\mathcal{S}_{\text{ref}}$ of $M = 100$ states and report soft outputs: channel probabilities $\pi^{\text{ch}} \in \Delta^C$, mean airtime budget $\mathbf{w} \in \Delta^3$ and normalised power. A zone controller averages the outputs of its APs (the *zone teacher*) and the venue controller averages the zone teachers into a consensus $(\bar{\pi}^{\text{ch}}, \bar{\mathbf{w}}, \bar{p})$; this average is the closed-form minimiser of $\frac{1}{Z}\sum_z [\mathrm{KL}(\pi_z^{\text{ch}} \| \pi^{\text{ch}}) + \|\mathbf{w}_z - \mathbf{w}\|^2 + \|p_z - p\|^2]$. The consensus is returned to the zones, and every agent distils its base layers towards it with a proximal term that keeps it close to its own model,

$$\min_{\theta_i} \; \mathrm{KL}\big(\bar{\pi}^{\text{ch}} \,\|\, \pi_i^{\text{ch}}\big) + \|\bar{\mathbf{w}} - \mathbf{w}_i\|^2 + \|\bar{p} - p_i\|^2 + \frac{\mu}{2}\|\theta_i - \theta_i^{\text{before}}\|^2, \qquad \mu = 0.01, \tag{10}$$

solved with 20 Adam steps (learning rate $5 \cdot 10^{-4}$). The reference set contains states from one rollout of an independent proxy venue under random actions, observed through the current predictor; it does *not* contain any agent's own transitions, and it is refreshed every five global rounds.

The alternative that exchanges weights at this level (FedProx averaging of the zone models, $\theta^* = (\sum_z \theta_z + \mu\theta_{\text{prev}})/(Z + \mu)$) is evaluated as a comparator in Section 4.7.

Communication accounting. We count the bytes of every message (32-bit floats). Per global round the distillation exchange costs $2ZM(C{+}4) \cdot 4$ bytes between zones and venue controller, instead of $2Z{\cdot}270{,}080{\cdot}4$ bytes for weights; the one-off and refresh cost of shipping $\mathcal{S}_{\text{ref}}$ ($M \cdot 797 \cdot 4$ bytes per zone and refresh) is added to the inter-zone total. Measured totals are reported in Table 11. Zone-level traffic between APs and their zone controller is the

same as in weight-based schemes and is reported separately.

3.7. Deployment Mapping: Channel Switching and QoS Queues

Channel switching. The channel in the action is applied at the start of the interval and costs service. A switching AP announces the change with channel-switch announcements (CSA); clients that miss it re-associate. We model this as an outage of $t_{\mathrm{sw}} = 3$ s of the 300 s interval: the AP's users receive a $t_{\mathrm{sw}}/300$ smaller share of the interval and the packets queued during the outage see an extra delay of $t_{\mathrm{sw}}/2$ on average, which enters the latency, throughput, reward and QoS metrics. The value is an engineering assumption (CSA with a few beacon intervals plus re-association of non-compliant clients); Section 4.8 reports results for $t_{\mathrm{sw}} \in \{0, 3, 10\}$ s and for a policy that was trained without any switching cost.

QoS queues. Commodity APs implement QoS with the four 802.11e EDCA access categories (AC_VO, AC_VI, AC_BE, AC_BK), each with its own contention parameters (AIFSN, $\mathrm{CW}_{\min}$, $\mathrm{CW}_{\max}$, TXOP limit). We map VoIP, video and best-effort to AC_VO, AC_VI and AC_BE. EDCA provides *relative priority*, not a bandwidth split, so the simplex action $\mathbf{w}_i$ is interpreted as an *airtime budget per access category* that is enforced on top of EDCA with the per-AC airtime quota (airtime-fairness / airtime-weight) scheduler available in current AP software, while the EDCA parameters stay at the standard values. To make the action implementable we (i) quantise $\mathbf{w}_i$ to multiples of 1/16 with a minimum of 1/16 per category (largest-remainder rounding), which is the granularity of integer airtime weights, and (ii) treat the budgets as limits that are work-conserving: unused airtime of an absent

or satisfied category is redistributed to the others. The simulator applies exactly this quantisation and redistribution. Cross-vendor differences in how airtime quotas are exposed, and the presence of a fourth (background) category, are outside the scope of the evaluation; they only change how the three numbers are programmed, not the decision problem.

4. Experiments

4.1. Simulator

All experiments use a Python simulator of a transit terminal with $N = 30$ APs on a 6×5 grid (30 m spacing, 2 m position jitter) in $Z = 3$ zones ($i \bmod 3$). It is a quasi-static model: all quantities are computed once per 5-minute control interval, for the users present in that interval. Table 5 lists the parameters. The simulator was extended for this revision with an explicit contention model, a channel-switching cost and the EDCA-style airtime mapping of Section 3.7.

Radio. Path loss is $PL(d) = 46.4 + 30 \log_{10} d$ dB (5 GHz, indoor), noise -95 dBm, and $C = 11$ orthogonal 20 MHz channels (adjacent-channel leakage is not modelled). A user associates to its nearest AP. The PHY rate of user u is $R_u = \min\{156,\ 0.75 \cdot 20 \log_2(1 + \mathrm{SINR}_u)\}$ Mbps (2 spatial streams, 256-QAM cap), with at least 1 Mbps.

MAC contention. Two APs that can hear each other (received power $\geq$ -82 dBm in either direction) and share a channel form a *carrier-sense domain*: they time-share the medium, each receiving the airtime fraction $n_i / \sum_{j \in \mathcal{D}_i} n_j$ of the domain's users, and they do not interfere with each other.

Algorithm 1 STAG-FedSAC training (one environment step shown; all APs act simultaneously).

Initialise ST-GCAT ϕ; for every AP i: actor π_i, critics Q_i, targets $\bar{Q}_i$, buffer $\mathcal{B}_i$, $\lambda_i = 0$.
for episode $= 1, \ldots, E$ **do**
    $o \leftarrow$ env.reset(); clear history
    for step $t = 1, \ldots, 12$ **do**
        Build $\mathbf{H}, \mathbf{S}, \mathbf{A}$; $\mathbf{Z}_{\text{fused}}, \hat{L} \leftarrow$ ST-GCAT$(\mathbf{H}, \mathbf{S}, \mathbf{A})$
        For every AP: $\mathbf{s}_i \leftarrow$ Eq. (5); $\mathbf{a}_i \sim \pi_i(\cdot|\mathbf{s}_i)$ (random during warm-up)
        $o', \{r_i\}, \text{done} \leftarrow$ env.step$(\{\mathbf{a}_i\})$
        $\mathcal{B}_i \leftarrow \mathcal{B}_i \cup \{(\mathbf{s}_i, \mathbf{a}_i, R_{\text{s}}(r_i + \delta_p b_i^{\text{pred}} - \text{penalty}_i), \mathbf{s}_i')\}$; update λ_i (Eq. 2)
        Train ST-GCAT on stored samples (Eq. 8; with the actor term only after warm-up)
        if step $\geq T_{\text{warmup}}$ **then**
            SAC update of critics, actors, temperature of all agents (Eqs. 6–7)
        end if
        if step $\bmod T_{\text{local}} = 0$ **then**
            Zone aggregation (Eq. 9)
        end if
        if step $\bmod T_{\text{global}} = 0$ **then**
            Proximal federated distillation (Eq. 10)
        end if
    end for
    Attach true multi-step targets to the episode's predictor samples
end for

Table 5: Simulator parameters.

Item	**Value**
Venue	30 APs, 6×5 grid, 30 m spacing, 3 zones
Path loss / noise / carrier-sense threshold	$46.4 + 30 \log_{10} d$ dB / -95 dBm / -82 dBm
Channels, bandwidth	11 orthogonal, 20 MHz
PHY rate	$\min\{156, 0.75 \cdot 20 \log_2(1 + \text{SINR})\}$ Mbps
MAC	Bianchi DCF, 16-MPDU A-MPDU, $\text{CW}_{\min}$ 16, 6 stages; 25
Offered rate VoIP / video / best-effort	0.1 / 3 / 1 Mbps per user (mix 20 / 30 / 50%)
SINR threshold VoIP / video / best-effort	10 / 15 / 5 dB; latency SLA 50 / 100 / 500 ms
Users	500–1000 at episode start; surge users stay two intervals (a
Channel-switch outage	3 s per switch (sensitivity: 0, 10 s)
Airtime budget granularity	1/16, minimum 1/16 per access category
AP power model	6 W static + transmit power / 0.3

Only co-channel APs outside the domain add interference to the SINR (hidden terminals are not modelled). With $m_i = \max(1, \lfloor 0.25 \sum_{j \in \mathcal{D}_i} n_j \rceil)$ backlogged stations in the domain, the fraction of airtime that carries payload is the saturation efficiency $\eta(m_i)$ of the IEEE 802.11 DCF obtained from Bianchi's fixed-point model [27] (802.11ac timings, 16-MPDU A-MPDUs, $\mathrm{CW}_{\min} = 16$, 6 back-off stages; $\eta(1) = 0.87$, $\eta(10) = 0.70$, $\eta(50) = 0.56$, $\eta(100) = 0.50$). The airtime available to AP i is $\rho_i^{\mathrm{avail}} = \eta(m_i)\, n_i / \sum_{j \in \mathcal{D}_i} n_j$. User u of class c is offered a constant rate d_c (0.1, 3 and 1 Mbps for VoIP, video and best-effort) and obtains the part of it that fits into its class's airtime budget (Section 3.7, with redistribution of unused airtime).

Latency. The per-class latency is $5 + s_i\,[1 + \varrho/(2(1-\varrho))]$ ms, an M/G/1-type delay with the domain service time $s_i =$ (A-MPDU payload time)$/\eta(m_i)$ and $\varrho = \min(0.98,$ airtime needed by the class/airtime granted to it); if the class is overloaded ($\varrho > 0.98$) the delay rises linearly to the 500 ms buffer limit as ϱ grows to 1.48. A channel switch adds half of the outage time to the delay of the AP's users (Section 3.7). Latency in this model is therefore never a constant: it depends on the number of contending stations, the class budget and the load.

Users and surges. At the start of an episode between 500 and 1000 users are placed uniformly in the venue (20% VoIP, 30% video, 50% best-effort) and move by a 2 m random walk per interval. An episode is 12 intervals (1 hour) and contains 3–7 timetable events (departure or arrival, 50–300 passengers, random zone and time). Around each event passengers join the venue for two intervals, at most 100 per interval, producing the load surges that the schedule announces. Timetable data are synthetic.

Metrics. Reward is the unscaled Eq. (1) without b^{pred} averaged over APs and intervals. *Throughput* is the mean per-user throughput (Mbps), *latency* the mean per-user latency, *JFI* Jain's index of the user satisfaction ratios ς_u over all users of the venue, *QoS satisfaction* the fraction of users that meet their class requirement (SINR, latency and $\varsigma_u \geq 0.5$), *energy efficiency* the total throughput divided by the power drawn by all APs (6 W static plus transmit power divided by a PA efficiency of 0.3), and *switches* the fraction of APs that change channel per interval. Forecast quality is measured by RMSE and MAE over all $\Delta = 6$ horizons (and for $h = 1$ only), MAPE on loads above 0.05, and *Surge-RMSE*: the RMSE over AP/interval pairs that lie within one interval of a scheduled event with at least 100 passengers in the AP's zone (Normal-RMSE: all other pairs).

4.2. Compared Methods

- **SSF** (strongest signal first, static): users keep their strongest AP, fixed round-robin channel plan, mid power, equal airtime split.
- **LLF** (least loaded first, reactive): lowers power on overloaded APs and raises it on idle ones, splits airtime by the observed QoS mix, fixed channel plan.
- **LSTM-DRL**: an LSTM forecasts the next loads of each AP from its own history (no graph, no schedule); one DDPG agent *shared by all APs* acts on the local features, the QoS mix and the scalar one-step forecast. No federation.
- **FedDDPG** (adapted from [2]): one DDPG agent per AP, flat FedAvg of actor and critic every T_{local} steps, no forecast, no schedule.

- **STAG-FedSAC** and its variants: *Proposed* zeroes the latent embedding $\mathbf{Z}_{\text{fused}}$ in the policy state (the policy uses the compact schedule features; the predictor is trained on the forecast error only, since the actor term has no effect on a zeroed embedding); *Latent Emb.,* η=0.1 feeds the embedding and uses the joint gradient; *Latent Emb.,* η=0 (A1) removes the joint gradient; *w/o Schedule* (A2) is the latent-embedding configuration with $\eta = 0.1$ and the schedule removed from the predictor and the state. The proposed configuration corresponds to ablation A3 (no embedding) and was singled out after the results were available.

- **Federation variants** (Section 4.7): isolated agents, flat FedAvg, zone FedAvg only (Level 1), zone FedAvg with weight-based global FedProx, and a pooled-experience learner.

- **Persistence**: a forecast reference that predicts the current load for all horizons (with SSF control).

All learned methods are trained with the same protocol (same simulator, the same reward plus Lagrangian penalty and reward scale, 200 warm-up episodes of exploration around the incumbent configuration, replay buffer, one gradient update per environment step, same number of episodes) and are evaluated on the same held-out episodes. The two DDPG baselines use Gaussian exploration noise $\sigma = 0.2$ and encode the channel as one continuous output rounded to an index. We did not tune any method separately; hyperparameters are shared wherever the architectures allow it.

4.3. Protocol and Hyperparameters

Every learned method is trained for 1000 episodes (the first 200 are warm-up) and evaluated on 20 held-out test episodes whose seeds are disjoint from the training and validation seeds and identical for all methods (Table 7: seeds 0–2). The first version of this paper stated 2000 training episodes; we use 1000 here because the learning curves flatten well before that point (Fig. 2) and because every method had to be re-run with the extended simulator. The federation variants of Section 4.7 use seeds 0–1 (STAG-FedSAC: 0–2), the robustness runs trained without contention or without switching cost seeds 0–1, the reward-weight and the η sensitivity runs seed 0 (weights) and seeds 0–2 ($\eta = 0.01$). The number of seeds is given in every table. A seed fixes the network initialisation, the simulator layout and the training episodes; the metrics of a run are the mean over its 20 test episodes, and the tables show the mean and standard deviation of these run-level values over seeds. A five-episode validation during training (seeds disjoint from the test seeds) produces the learning curves and was used for the only tuned value, κ.

4.4. Main Results

Overall Performance and Comparison with Baselines. Table 7 presents the primary evaluation results across all compared methods on held-out test episodes. STAG-FedSAC (Proposed) reaches a reward of 0.182 ± 0.006, higher than the reactive heuristic LLF (0.156) and the learned DRL baselines, and equal to that of the static operator incumbent plan SSF (0.181 ± 0.001) within seed variation (paired difference $+0.000$, 95% interval $[-0.004, +0.007]$; Table 8).

Table 6: Hyperparameters (identical for all learned methods where applicable); the incumbent configuration is round-robin channels, 14 dBm, equal airtime.

Parameter	**Symbol**	**Value**
Environment		
APs / channels / zones	N, C, Z	30, 11, 3
Control interval / episode		5 min / 12 intervals
History window / forecast horizon	T, Δ	12 (60 min) / 6 (30 min)
Channel-switch outage / airtime granularity	t_{sw} / EDCA levels	3 s / 1/16
Reward		
Throughput, latency, fairness, QoS, prediction	$\alpha, \beta, \gamma, \delta, \delta_p$	0.40, 0.25, 0.20, 0.05, 0.10
Reward scale for learning	R_{s}	10
ST-GCAT		
GAT / Transformer heads, layers, d_h	K, L, d_h	4, 2, 128
Predictor learning rate, batch	η_ϕ	5×10^{-4}, 16 samples
Joint gradient weight	η	0.10
SAC		
Actor / critic hidden layers		[256, 256] (+128 personal head)
Batch, replay size per AP		256, 24,000 transitions (all episo
Learning rate actor, critic, temperature		3×10^{-4}
Discount / soft target update	γ, τ	0.99 / 0.005
Initial temperature / target entropy		0.05 / -3
Gumbel-softmax temperature		$2.0 \rightarrow 0.5$ (linear)
Embedding scale in the state	κ	0.03 (chosen from $\{1, 0.125, 0.03$
HierFed-KD		
Zone / global round	T_ℓ, T_g	100 / 500 steps
Proximal weight, quality temperature	μ, τ_w	0.01, 0.1
Reference states, distillation steps, lr	M	100, 20, 5×10^{-4}
Constraint		
Utilisation limit, step, bound	$u_{\max}, \rho_\lambda, \lambda_{\max}$	0.9, 0.01, 1

Table 7: Held-out performance (20 test episodes per run; mean and standard deviation over training seeds). Bold: best. The reward is the unscaled Eq. (1) without b^{pred}. "Sw./AP/step" includes the initial reconfiguration from the random channels at the start of each episode, which every method pays equally.

Method	Seeds	Reward	Thrpt. (Mbps/user)	Lat. (ms)	JFI	QoS sat.	Mbps/W	Sw./AP/step
SSF (Static)	3	$0.181_{\pm 0.001}$	$1.31_{\pm 0.00}$	$124.6_{\pm 2.2}$	$\mathbf{0.985}_{\pm \mathbf{0.001}}$	$0.682_{\pm 0.002}$	$7.40_{\pm 0.03}$	$\mathbf{0.075}_{\pm \mathbf{0.000}}$
LLF (Reactive)	3	$0.156_{\pm 0.001}$	$1.22_{\pm 0.01}$	$145.0_{\pm 2.2}$	$0.967_{\pm 0.002}$	$0.638_{\pm 0.002}$	$6.85_{\pm 0.04}$	$\mathbf{0.075}_{\pm \mathbf{0.000}}$
LSTM-DRL	3	$0.108_{\pm 0.004}$	$1.11_{\pm 0.01}$	$198.9_{\pm 8.0}$	$0.928_{\pm 0.007}$	$0.577_{\pm 0.021}$	$6.07_{\pm 0.07}$	$0.668_{\pm 0.011}$
FedDDPG	3	$0.116_{\pm 0.004}$	$1.16_{\pm 0.01}$	$184.8_{\pm 7.2}$	$0.943_{\pm 0.005}$	$0.602_{\pm 0.010}$	$6.40_{\pm 0.08}$	$0.752_{\pm 0.031}$
STAG-FedSAC w/o Schedule	3	$0.128_{\pm 0.010}$	$1.16_{\pm 0.03}$	$183.9_{\pm 14.3}$	$0.937_{\pm 0.007}$	$0.584_{\pm 0.019}$	$6.45_{\pm 0.18}$	$0.214_{\pm 0.013}$
STAG-FedSAC (Latent Emb., η=0.1)	3	$0.141_{\pm 0.008}$	$1.19_{\pm 0.02}$	$170.6_{\pm 8.6}$	$0.952_{\pm 0.007}$	$0.614_{\pm 0.017}$	$6.65_{\pm 0.14}$	$0.165_{\pm 0.009}$
STAG-FedSAC (Latent Emb., η=0)	3	$0.176_{\pm 0.009}$	$1.30_{\pm 0.03}$	$118.4_{\pm 9.8}$	$0.979_{\pm 0.008}$	$0.679_{\pm 0.017}$	$7.34_{\pm 0.16}$	$0.128_{\pm 0.011}$
STAG-FedSAC (Proposed)	3	$\mathbf{0.182}_{\pm \mathbf{0.006}}$	$\mathbf{1.32}_{\pm \mathbf{0.02}}$	$\mathbf{109.9}_{\pm \mathbf{7.0}}$	$0.985_{\pm 0.005}$	$\mathbf{0.689}_{\pm \mathbf{0.014}}$	$\mathbf{7.47}_{\pm \mathbf{0.13}}$	$0.153_{\pm 0.014}$

In terms of user experience, STAG-FedSAC reduces the average user latency to 109.9 ± 7.0 ms. This is an 11.8% latency reduction compared to SSF (124.6 ± 2.2 ms; lower than SSF in each of the three seeds) and a 24.2% reduction compared to the reactive heuristic LLF (145.0 ± 2.2 ms). The mean per-user throughput (1.32 ± 0.02 Mbps/user against 1.31 for SSF), the QoS satisfaction (0.689 ± 0.014 against 0.682), the energy efficiency (7.47 ± 0.13 Mbps/W against 7.40) and the fairness (0.985, equal to SSF) are on par with SSF; their differences from SSF are within the seed-to-seed standard deviations.

Learned DRL Baselines and Stability Analysis. STAG-FedSAC outperforms the learned DRL baselines: it improves reward by +56.9% over FedDDPG (0.116) and by +68.5% over LSTM-DRL (0.108), while cutting latency by 40.5% (from 184.8 ms) and 44.7% (from 198.9 ms), respectively. Table 8 reports the paired difference statistics across matched random seeds; all paired differences against the learned baselines are strictly positive with bootstrap

Table 8: Paired difference in reward between STAG-FedSAC and each compared method over matched random seeds (positive: STAG-FedSAC better), 95% bootstrap confidence interval of the mean difference, exact two-sided sign-flip permutation p-value, and Bonferroni-adjusted p-value. The "w/o Schedule" model also uses the latent embedding, so its comparison with the proposed configuration changes two factors.

STAG-FedSAC vs.	Δ reward	95% CI	p	p_{Bonf}	Seeds
SSF (Static)	+0.000	[-0.004, +0.007]	1.000	1.000	3
LLF (Reactive)	+0.026	[+0.020, +0.032]	0.250	1.000	3
LSTM-DRL	+0.074	[+0.069, +0.078]	0.250	1.000	3
FedDDPG	+0.066	[+0.063, +0.068]	0.250	1.000	3
STAG-FedSAC w/o Schedule	+0.054	[+0.038, +0.069]	0.250	1.000	3
STAG-FedSAC (Latent Emb., η=0.1)	+0.041	[+0.031, +0.048]	0.250	1.000	3
STAG-FedSAC (Latent Emb., η=0)	+0.006	[+0.003, +0.008]	0.250	1.000	3

intervals entirely bounded away from zero.

A main factor in the poor performance of FedDDPG and LSTM-DRL appears to be *policy instability and frequent channel switching*. As shown in Table 7, FedDDPG and LSTM-DRL change the operating channel of a fraction 0.752 and 0.668 of the APs per control interval, respectively. In a realistic deployment with MAC contention and channel-switch outages (Section 3.7), frequent channel changes incur severe service interruption ($t_{\mathrm{sw}} = 3$ s) and queuing delays that reduce user throughput and increase latency. A plausible explanation, which we did not test directly, is that the DDPG baselines lack entropy regularisation and proximal constraints, so that policy updates drift away from the coordinated incumbent frequency plan.

In contrast, STAG-FedSAC changes far fewer channels (0.153 switches/AP/step, about twice the 0.075 of the static plan, which only pays the initial reconfiguration). We attribute this to properties of SAC (entropy regularisation, personal heads) and to the initialisation at the incumbent configuration, but we did not run an experiment that isolates these factors, and we did not analyse which power or airtime adjustments the learned policy makes.

Figure 2 illustrates the convergence trajectories across training episodes. Both DDPG baselines rapidly degrade immediately after the exploration warm-up phase (200 episodes), falling into high-latency sub-optimal policies. In contrast, the validation reward of STAG-FedSAC stays close to the level of the incumbent plan throughout training, without a clear upward trend (the latent-embedding configuration with $\eta = 0.1$ declines, Section 4.6). A perfect-information greedy optimiser (not deployable; no switching cost) evaluated on test episodes reaches a reward of 0.216 (+18% above SSF), so

headroom exists, but the learned controllers realise little of it.

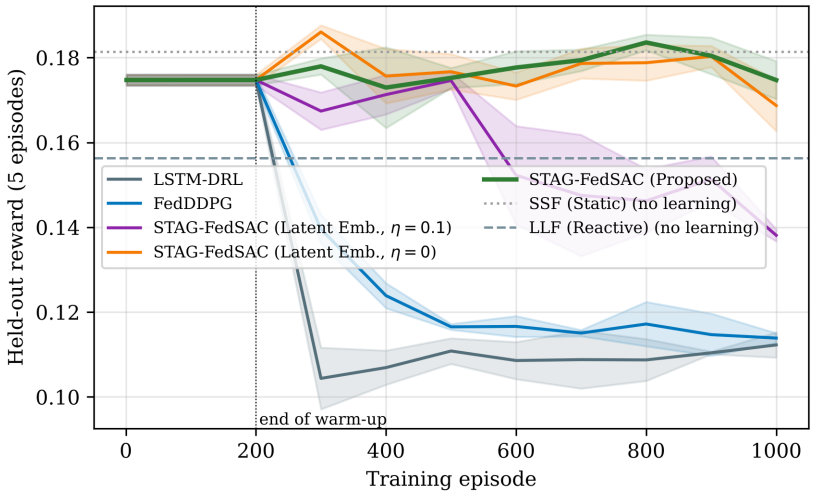


Figure 2: Reward on five held-out validation episodes during training (mean and standard deviation over seeds). During the first 200 episodes the actors only explore around the incumbent configuration (no gradient updates of the policy), so every learned method starts at the level of the incumbent plan; horizontal lines are the non-learning baselines.

4.5. Forecast Quality

Accurate multi-horizon load forecasting is the critical enabler of proactive radio resource allocation, providing the necessary lead time for APs to coordinate before crowds assemble. Table 9 and Figure 3 summarize the forecasting performance across all horizons ($\Delta = 6$, spanning 30 minutes).

ST-GCAT achieves an overall multi-step RMSE of 0.121, outperforming the persistence baseline (0.186, a 35% error reduction), the temporal-only LSTM baseline (0.142, 15%), and the schedule-unaware ST-GCAT model (0.175, 31%); these values are those of the decoupled ($\eta = 0$) row of Table 9, and the predictor of the proposed configuration, which has the same archi-

Table 9: Load forecast error on the held-out episodes (load is normalised to $[0, 1]$; all $\Delta = 6$ horizons unless stated). Surge: AP/interval pairs within one interval of a scheduled event with at least 100 passengers in the AP's zone.

Forecaster	RMSE	MAE	RMSE (h=1)	Surge-RMSE	Normal-RMSE	MAPE (%)
Persistence	$0.186_{\pm 0.001}$	$0.130_{\pm 0.000}$	$0.096_{\pm 0.000}$	$0.266_{\pm 0.002}$	$0.162_{\pm 0.001}$	$27.8_{\pm 0.4}$
LSTM (temporal only)	$0.142_{\pm 0.000}$	$0.107_{\pm 0.002}$	$0.075_{\pm 0.001}$	$0.213_{\pm 0.010}$	$0.118_{\pm 0.004}$	$24.4_{\pm 1.0}$
ST-GCAT w/o schedule	$0.175_{\pm 0.003}$	$0.135_{\pm 0.004}$	$0.159_{\pm 0.004}$	$0.243_{\pm 0.005}$	$0.152_{\pm 0.005}$	$30.2_{\pm 1.7}$
ST-GCAT (joint gradient, η=0.1)	$0.127_{\pm 0.000}$	$0.097_{\pm 0.001}$	$0.132_{\pm 0.000}$	$0.169_{\pm 0.006}$	$0.116_{\pm 0.002}$	$21.3_{\pm 0.6}$
ST-GCAT (decoupled, η=0)	$0.121_{\pm 0.000}$	$0.091_{\pm 0.001}$	$0.126_{\pm 0.003}$	$0.163_{\pm 0.009}$	$0.111_{\pm 0.002}$	$20.4_{\pm 0.7}$

tecture, reaches 0.121. The predictive advantage of ST-GCAT is particularly pronounced during critical scheduled surge windows: ST-GCAT achieves a Surge-RMSE of 0.163, representing a 39% reduction compared to persistence (0.266), 24% compared to the LSTM baseline (0.213), and 33% compared to the schedule-unaware model (0.243). Outside of surge windows, the LSTM and the ST-GCAT variants that use the schedule have a Normal-RMSE of 0.111–0.118, against 0.162 for persistence and 0.152 for ST-GCAT without schedule.

These results reveal two fundamental insights into spatio-temporal network forecasting:

1. **The Essential Role of Schedule Cross-Attention**: Without schedule conditioning, the graph-attention architecture alone achieves an RMSE of 0.175 and Surge-RMSE of 0.243. This indicates that, in our simulator (where the timetable is exact), the cross-attention over upcoming timetable metadata is the main source of the gain, and that the graph and temporal layers alone do not anticipate announced crowd arrivals from historical trends.

2. **Horizon-Adaptive Dynamics**: At the immediate one-step horizon ($h = 1$, 5 minutes ahead), historical load autocorrelation dominates: the RMSE is 0.126 for ST-GCAT against 0.096 for persistence and 0.075 for the LSTM, i.e. ST-GCAT is *less* accurate at $h = 1$. Its lower overall RMSE therefore comes from the longer horizons, where the timetable carries information that the history does not; we did not break the error down per horizon.

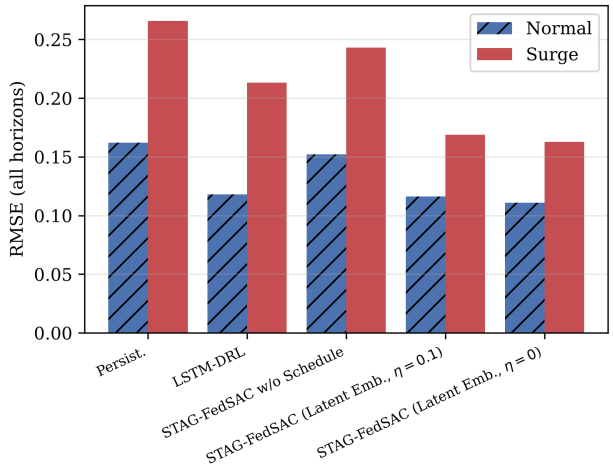


Figure 3: Forecast RMSE on surge and non-surge AP/interval pairs.

4.6. Ablation of the Contributions

To isolate the specific contributions of schedule conditioning, state representation design, and gradient coupling, we conduct systematic ablation studies (Tables 7, 9, and 10).

Impact of Schedule Conditioning (Ablation A2). Removing the transit schedule features from the predictor and the policy input degrades forecasting

clearly and control less clearly. The forecast RMSE rises from 0.121 to 0.175 and the surge-window RMSE from 0.161 to 0.243 (Table 9). For control, the like-for-like comparison is with the latent-embedding configuration with $\eta = 0.1$, which also uses the embedding: its reward falls from 0.141 to 0.128 (about 9%) and its latency rises from 170.6 ms to 183.9 ms (7.8%), but with three seeds the reward difference is within seed variation (its 95% interval includes zero). Against the proposed configuration the gap is larger (reward 0.182 to 0.128, latency 109.9 ms to 183.9 ms), but that comparison also changes the state representation. We read these results as evidence that the timetable carries information that the historical counters do not, for forecasting; for control the evidence is suggestive only.

State Representation Design: Direct Schedule Features vs. Latent Embeddings. We investigate how predictive schedule knowledge is best communicated to the DRL policy: via compact, explicit schedule features $\mathbf{S}_i^{t+1} \in \mathbb{R}^8$ versus the full high-dimensional spatio-temporal latent embedding $\mathbf{Z}_{\text{fused}} \in \mathbb{R}^{768}$. The results suggest the following for our setting:

- **Compact Direct Representation (Proposed STAG-FedSAC)**: Directly injecting the 8 normalized schedule features (event times, passenger counts, and surge indicators) into the SAC state provides a compact, stationary, and semantically invariant representation. Because these features depend only on the deterministic timetable, transitions stored in the off-policy replay buffer $\mathcal{B}_i$ remain valid across subsequent training iterations, which should favour stable Q-function estimation; this configuration gives the best control performance among the SAC variants (0.182, latency 109.9 ms).

- **Latent Embedding Injection**: In contrast, passing the uncompressed 768-dimensional latent embedding $\mathbf{Z}_{\text{fused}}$ into the policy state exposes off-policy training to *representation staleness*. As the ST-GCAT predictor updates its parameters ϕ during training, historical transitions stored in the replay buffer contain stale representations generated by earlier model checkpoints. A plausible consequence is non-stationarity in the critic's value estimates; we did not test this directly (for instance by recomputing the embeddings of stored transitions). The observed reward is lower than with the compact representation: 0.176 with $\eta = 0$ (latency 118.4 ms) and 0.141 with $\eta = 0.1$ (latency 170.6 ms), against 0.182 (109.9 ms).

In our experiments, compact schedule features worked better than concatenating the latent embedding; we did not test other ways of using the embedding.

Predictor–Policy Gradient Coupling (η Sweep). We examine the effect of the bidirectional joint gradient coupling weight η in Eq. (8), which backpropagates the actor objective into the ST-GCAT predictor. Table 10 evaluates $\eta \in \{0, 0.01, 0.10\}$. Among the three values tested, the decoupled objective ($\eta = 0$) gives the highest reward (0.176) and the lowest forecast error (RMSE 0.121, Surge-RMSE 0.163). Increasing the joint gradient weight to $\eta = 0.01$ and $\eta = 0.10$ lowers the control reward (to 0.146 and 0.141) and slightly worsens the forecast accuracy (RMSE 0.122 and 0.127).

A plausible explanation, which we did not test, is that in off-policy learning the critic evaluates historical buffer samples while the actor objective changes the predictor, so that backpropagating the policy gradient into ST-

GCAT perturbs the embeddings that the stored transitions were computed with. We could not show a benefit of the joint gradient at any of the three weights.

Table 10: Sensitivity to the joint gradient coupling weight η (mean $\pm$ standard deviation over seeds; $\eta = 0$ corresponds to decoupled training).

η	Seeds	Reward	Lat. (ms)	RMSE	Surge-RMSE
0	3	$0.176_{\pm 0.009}$	$118.4_{\pm 9.8}$	$0.121_{\pm 0.000}$	$0.163_{\pm 0.009}$
0.01	3	$0.146_{\pm 0.005}$	$164.0_{\pm 13.2}$	$0.122_{\pm 0.001}$	$0.167_{\pm 0.007}$
0.1	3	$0.141_{\pm 0.008}$	$170.6_{\pm 8.6}$	$0.127_{\pm 0.000}$	$0.169_{\pm 0.006}$

4.7. What Does Federation Buy?

Table 11: Federation schemes for the same agents. Traffic is measured during training, summed over all APs and rounds (MB); inter-zone traffic includes the reference states shipped for distillation.

Scheme	Seeds	Reward	Lat. (ms)	JFI	QoS sat.	Intra-zone (MB)	Inter-zone (MB)
Isolated (no federation)	2	$0.121_{\pm 0.004}$	$184.6_{\pm 2.8}$	$0.935_{\pm 0.009}$	$0.584_{\pm 0.015}$	0.0	0.0
Flat FedAvg	2	$0.141_{\pm 0.006}$	$162.8_{\pm 2.9}$	$0.954_{\pm 0.002}$	$0.615_{\pm 0.000}$	0.0	7778.3
Zone FedAvg only	2	$0.132_{\pm 0.005}$	$176.9_{\pm 9.0}$	$0.946_{\pm 0.010}$	$0.606_{\pm 0.021}$	7778.3	0.0
Hier. FedAvg+FedProx (weights)	2	$0.146_{\pm 0.005}$	$156.9_{\pm 9.3}$	$0.959_{\pm 0.009}$	$0.637_{\pm 0.010}$	7778.3	155.6
HierFed-KD (Proposed distillation)	3	$0.141_{\pm 0.008}$	$170.6_{\pm 8.6}$	$0.952_{\pm 0.007}$	$0.614_{\pm 0.017}$	7785.5	4.5
Pooled experience (centralised)	2	$0.100_{\pm 0.013}$	$213.0_{\pm 7.0}$	$0.902_{\pm 0.028}$	$0.534_{\pm 0.028}$	0.0	0.0

All federation variants share identical per-AP SAC agents, reward formulations, simulation environments, and random seeds, differing solely in how model parameters and knowledge are shared across the network hierarchy. All variants in this comparison use the latent-embedding configuration with $\eta = 0.1$; the HierFed-KD row therefore corresponds to the "Latent Emb.,

η=0.1" row of Table 7 (reward 0.141), not to the proposed configuration (0.182). Table 11 and Figure 4 summarize their held-out evaluation performance alongside exact byte-level communication measurements collected across the entire training campaign.

Benefits of Federated Collaboration. Sharing learned representations across the network hierarchy provides clear performance benefits over isolated learning. Isolated agents operating without federation achieve a reward of 0.121 and an average latency of 184.6 ms. In contrast, all federated variants achieve higher rewards (0.132–0.146) and reduced latency (156.9–176.9 ms). By sharing base feature representations, federated APs benefit from the collective experience of neighbouring APs experiencing diverse congestion conditions, which suggests that sharing helps in this setting (two to three seeds per variant, so the differences among the federated variants are not resolved).

Failure of Monolithic Experience Pooling. To examine whether unconstrained centralized data pooling could serve as an empirical upper bound, we evaluate a centralized learner that samples transitions indiscriminately from a single pooled replay buffer across all APs. Crucially, this pooled-experience approach achieves a reward of only 0.100 and an elevated latency of 213.0 ms, which is worse than isolated learning (two seeds). Because individual APs occupy heterogeneous spatial positions with distinct channel assignments, local contention domains, and traffic mixes, naive experience pooling creates severe policy-distribution mismatch for each agent's critic. This suggests that, for AP-specific policies, pooling the transitions of other APs is not helpful; we did not test other pooling schemes.

Bandwidth Efficiency and Privacy Boundaries. While achieving competitive control performance, HierFed-KD provides dramatic communication reductions across the wide-area inter-zone network. Flat FedAvg requires exchanging full model weights directly with a central controller, consuming 7778.30 MB of wide-area traffic over training. Weight-based hierarchical federation reduces this to 155.57 MB by aggregating models at zone controllers before global synchronization.

HierFed-KD slashes wide-area inter-zone communication to only 4.55 MB total (3.8 MB for periodic public reference states and 0.72 MB for soft policy distributions). This is a measured 34× bandwidth reduction relative to weight-based hierarchical federation and a 1711× reduction relative to flat FedAvg. Furthermore, because only soft policy distributions on an unlabelled public reference set are transmitted across administrative zones, HierFed-KD keeps local replay transitions and per-user identifiers at the AP and exchanges no model weights between zones; this is data minimisation, not a formal privacy guarantee, since distilled outputs can still leak information.

4.8. Robustness to the Simulator Assumptions

Table 12: Reward / latency (ms) of the same trained policies evaluated under different assumptions. A policy trained without contention ("w/o MAC") or without switching cost is evaluated in the full simulator in the first column.

Policy	Full MAC, 3 s outage	No contention	No switch cost	10 s outage
SSF (Static) (trained w/ MAC)	$0.181_{\pm 0.001}$ / $125_{\pm 2}$	$0.222_{\pm 0.001}$ / $70_{\pm 1}$	$0.184_{\pm 0.001}$ / $124_{\pm 2}$	$0.177_{\pm 0.001}$ / $135_{\pm 2}$
LLF (Reactive) (trained w/ MAC)	$0.156_{\pm 0.001}$ / $145_{\pm 2}$	$0.179_{\pm 0.002}$ / $117_{\pm 3}$	$0.159_{\pm 0.001}$ / $144_{\pm 2}$	$0.152_{\pm 0.001}$ / $155_{\pm 2}$
FedDDPG (trained w/ MAC)	$0.116_{\pm 0.004}$ / $185_{\pm 7}$	$0.174_{\pm 0.002}$ / $106_{\pm 4}$	$0.136_{\pm 0.004}$ / $176_{\pm 8}$	$0.084_{\pm 0.003}$ / $270_{\pm 7}$
STAG-FedSAC (Latent Emb., η=0.1) (trained w/ MAC)	$0.141_{\pm 0.008}$ / $171_{\pm 9}$	$0.200_{\pm 0.005}$ / $93_{\pm 4}$	$0.146_{\pm 0.007}$ / $168_{\pm 8}$	$0.134_{\pm 0.008}$ / $190_{\pm 10}$
FedDDPG (trained w/o MAC)	$0.060_{\pm 0.000}$ / $257_{\pm 16}$	$0.165_{\pm 0.000}$ / $123_{\pm 9}$	$0.073_{\pm 0.001}$ / $251_{\pm 16}$	$0.041_{\pm 0.004}$ / $327_{\pm 4}$
STAG-FedSAC (Latent Emb., η=0.1) (trained w/o MAC)	$0.105_{\pm 0.010}$ / $217_{\pm 22}$	$0.178_{\pm 0.002}$ / $121_{\pm 5}$	$0.109_{\pm 0.009}$ / $216_{\pm 20}$	$0.099_{\pm 0.008}$ / $238_{\pm 17}$
STAG-FedSAC (Latent Emb., η=0.1) (trained w/o switch cost)	$0.134_{\pm 0.015}$ / $174_{\pm 22}$	$0.192_{\pm 0.007}$ / $100_{\pm 11}$	$0.139_{\pm 0.015}$ / $172_{\pm 21}$	$0.123_{\pm 0.016}$ / $202_{\pm 21}$

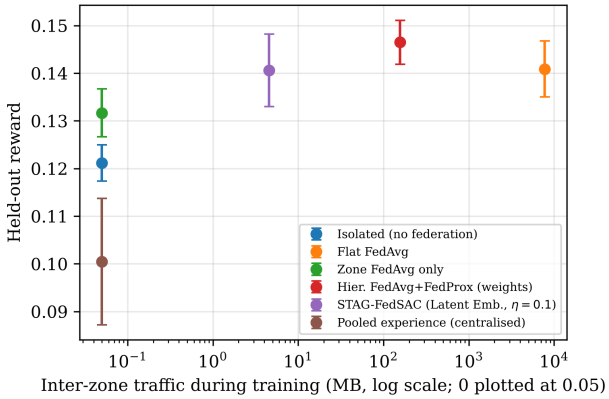


Figure 4: Held-out reward against inter-zone traffic during training for the federation schemes of Table 11 (mean and standard deviation over seeds; zero traffic is plotted at 0.05 MB).

To evaluate system resilience under varying physical and medium conditions, Table 12 reports performance across different MAC contention regimes and channel-switching outage durations. These experiments use the latent-embedding configuration with $\eta = 0.1$ ("STAG-FedSAC (latent)" below; reward 0.141 and latency 170.6 ms in Table 7), not the proposed configuration.

Impact of MAC-Layer Contention. Evaluating policies in an idealized contention-free setting yields artificially low latency: SSF latency drops from 124.6 ms to 70.3 ms and STAG-FedSAC (latent) latency drops from 170.6 ms to 92.5 ms (a factor of approximately 1.8×), while reward increases from 0.141 to 0.200. This gap indicates that, in our simulator, neglecting medium contention and queuing dynamics understates network latency by approximately 45%, which is why contention is modelled explicitly.

Crucially, policies *trained* without MAC contention awareness and subsequently evaluated in the realistic contention environment perform clearly

worse: STAG-FedSAC (latent) drops from 0.141 to 0.105 with latency spiking to 217 ms, while FedDDPG drops from 0.116 to 0.060 with latency escalating to 257 ms. This indicates that policies trained without contention are worse when contention is present (two training seeds; we did not inspect the learned plans).

Sensitivity to Channel-Switching Outages. Evaluating STAG-FedSAC (latent) across channel-switch outage durations $t_{\text{sw}} \in \{0, 3, 10\}$ s yields rewards of 0.146, 0.141, and 0.134, with corresponding latencies of 167.5, 170.6, and 189.6 ms. The switching penalty disproportionately impacts unstable policies: FedDDPG, which changes the channel of about 75% of the APs per interval, falls from 0.136 down to 0.084 as t_{sw} rises from 0 to 10 s, with latency escalating to 269.6 ms. STAG-FedSAC (latent), which changes 0.165 channels/AP/step, is less affected (reward 0.146 to 0.134 and latency 167.5 to 189.6 ms between 0 and 10 s), but not unaffected.

4.9. Sensitivity to the Reward Weights

Table 13: Sensitivity to the reward weights (one training seed per cell; "STAG-FedSAC" is the latent-embedding configuration with $\eta = 0.1$; reward is computed with the weights used in training, so rewards are not comparable across rows; latency and JFI are).

	SSF			FedDDPG			STAG-FedSAC (latent)		
Weights ($\alpha/\beta/\gamma/\delta$)	Reward	Lat.	JFI	Reward	Lat.	JFI	Reward	Lat.	JFI
Default (0.40/0.25/0.20/0.05)	0.182	124	0.986	0.114	189	0.941	0.133	180	0.947
Equal weights (0.25/0.25/0.25/0.25)	0.106	124	0.986	0.046	165	0.935	0.059	166	0.952
Throughput-heavy (0.55/0.15/0.20/0.10)	0.308	124	0.986	0.226	196	0.928	0.250	188	0.942
Latency-heavy (0.20/0.50/0.20/0.10)	-0.106	124	0.986	-0.182	186	0.936	-0.122	135	0.972

The reward weights are design choices (Section 3.2). To see whether the

conclusions depend on them, we retrained STAG-FedSAC (latent-embedding configuration, $\eta = 0.1$) and FedDDPG with three other weightings and evaluated SSF with the same weights (Table 13; the first row repeats the default weights for the same seed). The ordering of the methods is the same for all four weightings: SSF has the highest reward, STAG-FedSAC (latent) is second and FedDDPG third. STAG-FedSAC (latent) also has a higher JFI than FedDDPG in every row and a lower latency in three of the four rows (the equal-weights row is a tie within 2 ms), the largest gap being under the latency-heavy weights. Each cell is a single training seed, so the table indicates robustness of the qualitative picture, not precise differences.

5. Discussion and Practical Deployment Considerations

The experimental campaign provides several fundamental insights into the design, stability, and optimization of schedule-driven learning frameworks for high-density wireless venues:

Key Empirical Findings.

1. **Value of Schedule Anticipation**: Integrating transit timetables into spatio-temporal graph attention forecasting provides anticipatory lead time. ST-GCAT reduces the surge-window forecast error by 39% relative to persistence, by 33% relative to the same predictor without the schedule and by 24% relative to a temporal-only LSTM. For control the evidence is weaker: removing the schedule from the latent-embedding configuration lowers its reward by about 9% and raises its latency by 7.8%, within seed variation; against the proposed configuration the gap is larger (reward −30%, latency +67%), but that comparison also

changes the state representation. Timetable awareness is thus clearly beneficial for forecasting and suggestive for control.

2. **Representation Stationarity in Wireless DRL**: In our experiments, conditioning the policy directly on compact, physical schedule features worked better than concatenating the high-dimensional latent embedding of a concurrently updating predictor. Representation drift and staleness of embeddings in the replay buffer is a plausible but untested explanation. Decoupled training ($\eta = 0$) gave the best forecast error among the η values tested and the best control reward among the configurations that use the embedding, but the proposed configuration, which does not use the embedding, had a higher control reward still.
3. **Protocol and Medium Realism**: Accounting for IEEE 802.11 MAC contention, collision domains, and channel-switching penalties changes the evaluation materially. Neglecting contention underestimates latency by roughly 45%, and policies trained without contention perform worse when evaluated with it. The standard DRL baselines (FedDDPG, LSTM-DRL) change the channel of 0.67–0.75 of the APs per interval, against 0.15 for STAG-FedSAC (and 0.075 for the static plan); we attribute the difference to SAC's entropy regularisation, the personal heads and the incumbent initialisation, but did not isolate these factors.
4. **Data Minimisation and Bandwidth Efficiency**: By replacing raw parameter sharing across administrative zones with proximal knowledge distillation over soft policy outputs on a public reference set, HierFed-KD reduces inter-zone wide-area traffic by 34× relative to

weight-based hierarchical aggregation and 1711× relative to flat FedAvg, so that local replay transitions and association records are not transmitted (data minimisation, not a formal privacy guarantee).

Practical Deployment Considerations. Translating STAG-FedSAC into commercial WLAN architectures involves several practical considerations:

- **AP Hardware Interfaces and EDCA Quotas**: STAG-FedSAC's action space is designed to be realisable with commodity enterprise AP software; this has not been validated on hardware. The discrete channel choice is applied via standard 802.11 Channel Switch Announcements (CSA) to allow smooth client channel migration. The multi-class airtime split is enforced at standard 1/16 integer quantization via the per-AC airtime quota / airtime fairness scheduler present in contemporary enterprise Linux drivers (e.g., mac80211 airtime weight framework).

- **Schedule Uncertainty and Real-World Delays**: In operational transit hubs, scheduled flights and trains can experience gate reassignments, departure delays, or cancellations. In this work, timetables were evaluated under synthetic schedules. In production deployments, real-time timetable APIs (e.g., airport Flight Information Display Systems) can feed dynamic schedule updates directly into the ST-GCAT cross-attention queries. Extending ST-GCAT to handle probabilistic timetable confidence intervals represents an important operational extension.

- **Administrative Boundaries and Data Minimization**: In multi-tenant transit venues (e.g., terminal concourses vs. airline-operated

VIP lounges), separate administrative entities manage subsets of APs. HierFed-KD exchanges only soft policy outputs over an unlabelled public reference set generated from an independent proxy domain, so that raw user association histories and proprietary network traffic traces are not sent across administrative boundaries (the reference states and the soft outputs are).

Limitations. All results are from simulation and are quasi-static (one evaluation per 5-minute interval). The simulator is not calibrated to measurements, uses Bianchi's saturation approximation instead of packet-level MAC dynamics, and does not model hidden terminals, adjacent-channel interference or client roaming. The timetable is synthetic and exact. The 3 s switching outage and the EDCA airtime mapping are assumptions that were not validated on hardware. The main comparison uses three training seeds and 1000 episodes; with three seeds no sign-flip test can give $p < 0.25$, so we rely on effect sizes and bootstrap intervals. The proposed configuration was singled out among the ablation variants after the results were available, and the design choices made during development (reward scale, initial entropy temperature, update ratio, incumbent initialisation, embedding scale) were tuned on validation episodes mostly with variants of STAG-FedSAC, whereas the baselines were not tuned separately. The federation, robustness and reward-weight experiments use the latent-embedding configuration. In reward, the learned controllers match but do not clearly exceed the static incumbent plan, while a greedy perfect-information optimiser reaches 0.216 against 0.183. The federation provides data minimisation only: no differential privacy or secure aggregation is used, and the public reference set is assumed to be available

from an independent proxy venue.

Future Work. Key directions for future research include: (i) validating STAG-FedSAC on a physical multi-AP software-defined radio (SDR) testbed and high-fidelity packet-level simulators (e.g., ns-3 [28]) incorporating frame-level hidden terminal dynamics and 802.11ax/be trigger-based multi-user uplinks; (ii) incorporating formal differential privacy and secure multiparty aggregation into the inter-zone distillation protocol; and (iii) developing safe reinforcement learning mechanisms that enforce hard quality-of-service guarantees during early online adaptation.

6. Conclusion

In this paper, we introduced **STAG-FedSAC**, an end-to-end framework for proactive, schedule-driven WiFi radio resource management in high-density transit venues. STAG-FedSAC bridges predictive traffic forecasting and decentralized network control through three coordinated modules: (i) ST-GCAT, a schedule-conditioned spatio-temporal graph attention Transformer that predicts AP congestion up to 30 minutes ahead by cross-attending transit timetables with inter-AP interference topologies; (ii) per-AP Soft Actor-Critic agents that optimize transmit power, channel selection, and quantized 802.11e EDCA airtime quotas under a Lagrangian airtime utilization constraint; and (iii) HierFed-KD, a two-level hierarchical federated distillation protocol designed for multi-operator administrative boundaries.

Evaluated in a comprehensive simulation environment incorporating Bianchi's 802.11 MAC contention model, channel-switch outage penalties, and EDCA-based airtime budgeting, the proposed configuration of STAG-FedSAC matches

the static incumbent plan SSF in reward (0.182 against 0.181, within seed variation) and lowers the average user latency to 109.9 ms (11.8% below SSF at 124.6 ms), with throughput (1.32 Mbps), QoS satisfaction (0.689) and energy efficiency (7.47 Mbps/W) on par with SSF and 0.15 channel switches/AP/step (SSF: 0.075). Compared to learned DRL baselines (Fed-DDPG and LSTM-DRL), STAG-FedSAC improves reward by over 56% and cuts latency by more than 40%, with far fewer channel switches (0.15 against 0.67–0.75 switches/AP/step). Furthermore, ST-GCAT reduces the surge-window forecast RMSE by 39% relative to persistence and by 33% relative to the same predictor without the schedule, while HierFed-KD reduces inter-zone wide-area communication by 34× relative to weight-based hierarchical federation and 1711× relative to flat FedAvg. In our simulator, the ablation and sensitivity studies indicate that compact schedule features worked better than the latent embedding and that the joint predictor–policy gradient did not help; these findings are limited by the simulation-only evaluation and the small number of seeds.

Declaration of Competing Interest

The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No measured data are used. The simulator, the synthetic timetables, the run records of all experiments and the analysis scripts are included in the public repository `https://github.com/vinhqdang/STAG_FedSAC`.

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Ethics declaration

Not applicable.

References

[1] G. Kougioumtzidis, V. K. Poulkov, P. I. Lazaridis, Z. D. Zaharis, Mobile network traffic prediction using temporal fusion transformer, IEEE Transactions on Artificial Intelligence (2025).

[2] H. Du, Z. Li, D. Niyato, J. Kang, Z. Xiong, D. I. Kim, Enabling AI-generated content (AIGC) services in wireless edge networks, IEEE Wireless Communications 31 (1) (2024) 16–24. doi:10.1109/MWC.011.2200494.

[3] X. Pang, J. Hu, P. Sun, J. Ren, Z. Wang, When federated learning meets knowledge distillation, IEEE Wireless Communications 31 (5) (2024) 208–214.

[4] A. Alauthman, T. Shraa, Deep reinforcement learning-driven dynamic spectrum access in dense Wi-Fi environments, IEEE Access (2025).

[5] J. Yu, L. Liang, C. Guo, Z. Guo, S. Jin, G. Y. Li, Heterogeneous multi-agent reinforcement learning for distributed channel access in WLANs, IEEE Transactions on Cognitive Communications and Networking (2026).

[6] S. S. Shaabanzadeh, J. Sánchez-González, A spatio-temporal prediction methodology based on deep learning and real Wi-Fi measurements, Computer Networks 250 (2024) 110569. doi:10.1016/j.comnet.2024.110569.

[7] Z. Cai, C. Tan, J. Zhang, L. Zhu, Y. Feng, DBSTGNN-Att: Dual branch spatio-temporal graph neural network with an attention mechanism for cellular network traffic prediction, Applied Sciences 14 (5) (2024) 2173. doi:10.3390/app14052173.

[8] J. Lin, T. Lan, B. Zhang, et al., Multi-scenario cellular KPI prediction based on spatiotemporal graph neural network, IEEE Transactions on Automation Science and Engineering (2024). doi:10.1109/TASE.2024.3416912.

[9] J. Xiao, Y. Cong, W. Zhang, et al., A cellular traffic prediction method based on diffusion convolutional GRU and multi-head attention mechanism, Cluster Computing 28 (2025) 125. doi:10.1007/s10586-024-04848-y.

[10] S. Liu, M. He, Z. Wu, P. Lu, W. Gu, Spatial–temporal graph neural network traffic prediction based load balancing with reinforcement learning in cellular networks, Information Fusion 103 (2024) 102079. doi:10.1016/j.inffus.2023.102079.

[11] K. Yang, J. Yang, C. Shen, Average reward reinforcement learning for wireless radio resource management, in: 2024 58th Asilomar Conference on Signals, Systems, and Computers, IEEE, 2024, pp. 1188–1193.

[12] B. Adhikari, A. S. Khwaja, M. Jaseemuddin, A. Anpalagan, A. Nallanathan, Energy efficient RIS-assisted UAV networks using twin delayed DDPG technique, IEEE Transactions on Wireless Communications 23 (12) (2024) 18423–18439. doi:10.1109/TWC.2024.3468162.

[13] M. Wang, X. Liu, F. Wang, Y. Liu, T. Qiu, M. Jin, Spectrum-efficient user grouping and resource allocation based on deep reinforcement learning for mmWave massive MIMO-NOMA systems, Scientific Reports 14 (2024) 8884. doi:10.1038/s41598-024-59241-x.

[14] P. Pi, G. Ren, Soft actor–critic optimization for efficient NOMA uplink in intelligent vehicular networks, Physical Communication 68 (2024) 102581. doi:10.1016/j.phycom.2024.102581.

[15] K. Qi, Q. Wu, P. Fan, N. Cheng, W. Chen, J. Wang, K. B. Letaief, Deep-reinforcement-learning-based AoI-aware resource allocation for RIS-aided IoV networks, arXiv:2406.11245 (2024).

[16] S. M. Azimi-Abarghouyi, V. Fodor, Scalable hierarchical over-the-air

federated learning, IEEE Transactions on Wireless Communications 23 (2024). doi:10.1109/TWC.2024.3350923.

[17] M. F. Pervej, R. Jin, H. Dai, Hierarchical federated learning in wireless networks: Pruning tackles bandwidth scarcity and system heterogeneity, IEEE Transactions on Wireless Communications 23 (9) (2024) 11417–11432. doi:10.1109/TWC.2024.3396873.

[18] J. Shao, F. Wu, J. Zhang, Selective knowledge sharing for privacy-preserving federated distillation without a good teacher, Nature Communications 15 (2024) 349. doi:10.1038/s41467-023-44383-9.

[19] Y. Liang, Q. Chen, G. Zhu, H. Jiang, Y. C. Eldar, S. Cui, Communication-and-energy efficient over-the-air federated learning, IEEE Transactions on Wireless Communications 24 (1) (2025) 767–782. doi:10.1109/TWC.2024.3501297.

[20] B. Wu, F. Fang, X. Wang, Joint age-based client selection and resource allocation for communication-efficient federated learning over NOMA networks, IEEE Transactions on Communications 72 (1) (2024) 179–192. doi:10.1109/TCOMM.2023.3326774.

[21] H. Li, J. Chen, W. Xu, M. Liu, Knowledge distillation for asynchronous federated reinforcement learning in IoT, Knowledge-Based Systems 310 (2025) 113008. doi:10.1016/j.knosys.2025.113008.

[22] Y. Dai, L. Lyu, N. Cheng, M. Sheng, J. Liu, X. Wang, S. Cui, L. Cai, X. Shen, A survey of graph-based resource management in wireless networks — part II: Learning approaches, IEEE Transactions on Cognitive

Communications and Networking 11 (4) (2024). doi:10.1109/TCCN.2024.3508777.

[23] Z. Liu, J. Zhang, E. Shi, Z. Liu, D. Niyato, B. Ai, X. Shen, Graph neural network meets multi-agent reinforcement learning: Fundamentals, applications, and future directions, IEEE Wireless Communications 31 (2024) 39–47. doi:10.1109/MWC.015.2300595.

[24] M. Ji, Q. Wu, P. Fan, N. Cheng, W. Chen, J. Wang, K. B. Letaief, Graph neural networks and deep reinforcement learning based resource allocation for V2X communications, IEEE Internet of Things Journal (2024). doi:10.1109/JIOT.2024.3468162.

[25] M. Marwani, G. Kaddoum, Graph neural networks approach for joint wireless power control and spectrum allocation, IEEE Transactions on Machine Learning in Communications and Networking 2 (2024) 717–732. doi:10.1109/TMLCN.2024.3408723.

[26] B. Li, L.-L. Yang, R. G. Maunder, S. Sun, P. Xiao, Heterogeneous graph neural network for power allocation in multicarrier-division duplex cell-free massive MIMO systems, IEEE Transactions on Wireless Communications 23 (2) (2024) 962–977. doi:10.1109/TWC.2023.3325973.

[27] G. Bianchi, Performance analysis of the IEEE 802.11 distributed coordination function, IEEE Journal on Selected Areas in Communications 18 (3) (2000) 535–547. doi:10.1109/49.840210.

[28] ns-3 Consortium, ns3-ai: Reinforcement learning interface for ns-3

(2023).

URL https://github.com/hust-diangroup/ns3-ai